REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Position Time Graphs Practice Questions

Grade 9 Advanced | Module 2 Lesson 3 | Position Time Graphs

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Graph slope = Δx/Δt; straight-line model x = xi + vt. The graph shows position, not the physical shape of the path.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 37–41.

P01. The slope of a position–time graph represents:

A. Position only B. Elapsed time
C. Mass D. Velocity

P02. A horizontal position–time graph indicates:

A. Rest B. Constant positive speed
C. Constant negative speed D. Increasing acceleration

P03. A straight line with negative slope indicates:

A. Positive acceleration B. Constant velocity in the negative direction
C. Increasing position D. Zero velocity

P04. Two position–time curves intersect when the objects have:

A. The same velocity necessarily B. The same mass
C. The same position at the same time D. The same distance travelled necessarily

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P05. A steeper straight position–time line means:

A. Greater mass B. Longer elapsed time necessarily
C. Smaller velocity magnitude always D. Greater speed magnitude

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P06. A straight position–time graph joins (0 s, -2 m) and (6 s, 10 m). Calculate its slope.

A. 4 m/s B. 1 m/s
C. 2 m/s D. -2 m/s

P07. A straight position–time graph is described by x = 4 + 3t, in SI units. Find the position at t = 4 s.

A. 32 m B. 8 m
C. -16 m D. 16 m

P08. On position–time graphs, A follows xA = 3t and B remains at xB = 24 m. At what time do they meet?

A. 8 s B. 16 s
C. 4 s D. -8 s

P09. A position–time graph is horizontal from t = 2 s to t = 10 s. For how long is the object stationary?

A. 16 s B. 8 s
C. 4 s D. -8 s

P10. An object moves from -6 m to 12 m in 8 s. Calculate its average velocity.

A. 4.5 m/s B. 1.125 m/s
C. 2.25 m/s D. -2.25 m/s

P11. Which quantity belongs on each axis of a position–time graph?

P12. How is average velocity found between two graph points?

P13. What does a curved position–time graph indicate about velocity?

P14. Why is a position–time graph not a map of the object's path?

P15. What graph feature determines initial position at t = 0?

P16. A straight position–time graph joins (0 s, -3 m) and (9 s, 15 m). Calculate its slope.

P17. A straight position–time graph joins (0 s, -4 m) and (12 s, 20 m). Calculate its slope.

P18. A straight position–time graph is described by x = 6 + 3t, in SI units. Find the position at t = 5 s.

P19. A straight position–time graph is described by x = 8 + 3t, in SI units. Find the position at t = 6 s.

P20. On position–time graphs, A follows xA = 3t and B remains at xB = 36 m. At what time do they meet?

P21. On position–time graphs, A follows xA = 3t and B remains at xB = 48 m. At what time do they meet?

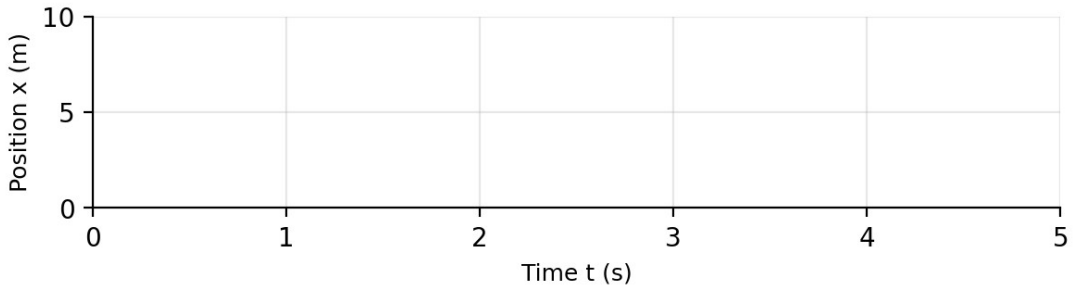
P22. A position–time graph is horizontal from t = 3 s to t = 15 s. For how long is the object stationary?

P23. A position–time graph is horizontal from t = 4 s to t = 20 s. For how long is the object stationary?

P24. An object moves from -9 m to 18 m in 12 s. Calculate its average velocity.

P25. An object moves from -12 m to 24 m in 16 s. Calculate its average velocity.

P26. Plot the data t = 0, 1, 2, 3, 4 s and x = −2, 0, 2, 4, 6 m. Describe the motion.



P27. Object A follows x = 2t and object B follows x = 12 − t, in SI units. Find when and where their graphs intersect.

P28. A graph rises, then becomes horizontal, then falls. Describe the motion without assuming numerical speeds.

P29. Can two objects have the same position but different velocities? Explain using graphs.

P30. How should you estimate a graph slope reliably from a best-fit straight line?